

5.1

$$(a) f(10, 10) = 0.08$$

$$(b) P(X_1 \geq 10, X_2 \leq 10) = f(10, 0) + f(10, 10) + f(20, 0) + f(20, 10) \\ = 0.10 + 0.08 + 0.05 + 0.05 = 0.28$$

$$(c) f_1(10) = \sum_{x_2} f(10, x_2) = f(10, 0) + f(10, 10) + f(10, 20) = 0.10 + 0.08 + \\ 0.12 \\ = 0.30$$

$$(d) f_2(20) = \sum_{x_1} f(x_1, 20) = f(5, 20) + f(10, 20) + f(20, 20) = 0.10 + 0.12 + \\ 0.18 \\ = 0.40$$

$$(e) f_1(x_1) = \begin{cases} 0.42, & x_1=5 \\ 0.30, & x_1=10 \\ 0.28, & x_1=20 \\ 0 & \text{otherwise} \end{cases} \quad f_2(x_2) = \begin{cases} 0.37, & x_2=0 \\ 0.23, & x_2=10 \\ 0.40, & x_2=20 \\ 0, & \text{otherwise} \end{cases}$$

$$(f) P(X_1 \geq 10) = f_1(10) + f_1(20) = 0.30 + 0.28 = 0.58$$

$$(g) P(X_2 \leq 100) = f_2(0) + f_2(10) = 0.37 + 0.23 = 0.60$$

5.3

(a)

$$f_1(x) = \sum_{y=1}^4 f(x, y) = \frac{4}{12} = \frac{1}{3}, \quad x = 1, 2, 3 \\ f_2(y) = \sum_{x=1}^3 f(x, y) = \frac{3}{12} = \frac{1}{4}, \quad y = 1, 2, 3, 4$$

(b) X and Y are independent because

$$f_1(x)f_2(y) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12} = f(x, y).$$

5.9

(a) For $x > 0$ and $y > 0$,

$$F(x, y) = (1 - e^{-2x})(1 - e^{-3y}) = F_1(x)F_2(y).$$

$$f_1(x) = F_1'(x) = \frac{d}{dx}(1 - e^{-2x}) = 2e^{-2x}, x > 0$$

$$f_2(y) = F_2'(y) = \frac{d}{dy}(1 - e^{-3y}) = 3e^{-3y}, y > 0$$

$$f(x, y) = f_1(x)f_2(y) = 6e^{-2x-3y}, x > 0, y > 0$$

Alternatively, it can be obtained as

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \frac{\partial}{\partial x}(3e^{-3y} - 3e^{-2x-3y}) = 6e^{-2x-3y}, x > 0, y > 0.$$

(b) From part (a), X and Y are independent.

(c) Because X and Y are independent, $f_1(x|y) = f_1(x) = 2e^{-2x}, x > 0$.

(d) $P(X \leq 2 \text{ and } Y \geq 3) = P(X \leq 2)P(Y \geq 3) = F_1(2)[1 - F_2(3)] = (1 - e^{-4})e^{-9} = e^{-9} - e^{-13}$

(e) X has exponential distribution with mean $1/2$ and Y has exponential distribution with mean $1/3$.

(f) $\text{Var}(X) = 1/4, \text{Var}(Y) = 1/9$

(g) Because X and Y are independent, the correlation coefficient is 0.

5.13

(a)

$$\begin{aligned} f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^3 \frac{4}{7} \left(x^2 + \frac{xy}{3} \right) dy = \left[\frac{4x^2 y}{7} + \frac{2xy^2}{21} \right]_{y=0}^3 \\ &= \frac{12x^2 + 6x}{7}, \quad 0 < x < 1 \end{aligned}$$

$$\begin{aligned} f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{4}{7} \left(x^2 + \frac{xy}{3} \right) dx = \left[\frac{4x^3}{21} + \frac{2x^2 y}{21} \right]_{x=0}^1 = \frac{2y + 4}{21}, \\ &0 < y < 3 \end{aligned}$$

(b)

$$F_1(x) = \begin{cases} 0 & x \leq 0 \\ \int_{-\infty}^x f_1(t) dt = \int_0^x \frac{12t^2 + 6t}{7} dt = \left[\frac{4t^3 + 3t^2}{7} \right]_0^x = \frac{4x^3 + 3x^2}{7}, & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

$$F_2(y) = \begin{cases} 0 & y \leq 0 \\ \int_{-\infty}^y f_2(t) dt = \int_0^y \frac{2t + 4}{21} dt = \left[\frac{t^2 + 4t}{21} \right]_0^y = \frac{y^2 + 4y}{21}, & 0 < y < 3 \\ 1 & y \geq 3 \end{cases}$$

(c)

$$P(Y < 2) = F_2(2) = \frac{12}{21} = \frac{4}{7}$$

(d)

$$\begin{aligned} P\left(X > \frac{1}{2}, Y < 1\right) &= \int_{-\infty}^1 \int_{1/2}^{\infty} f(x, y) dx dy = \int_0^1 \int_{1/2}^1 \frac{4}{7} \left(x^2 + \frac{xy}{3}\right) dx dy \\ &= \int_0^1 \left(\left[\frac{4x^3}{21} + \frac{2x^2 y}{21} \right]_{x=1/2}^1 \right) dy = \int_0^1 \left[\left(\frac{4 + 2y}{21} \right) - \left(\frac{1 + y}{42} \right) \right] dy \\ &= \int_0^1 \left(\frac{1}{6} + \frac{y}{14} \right) dy = \left[\frac{y}{6} + \frac{y^2}{28} \right]_0^1 = \frac{1}{6} + \frac{1}{28} = \frac{17}{84} \end{aligned}$$

(e)

$$f_2(y|x) = \frac{f(x, y)}{f_1(x)} = \frac{\frac{4}{7} \left(x^2 + \frac{xy}{3}\right)}{\frac{12x^2 + 6x}{7}} = \frac{2x \left(x + \frac{y}{3}\right)}{3x(2x + 1)} = \frac{2(3x + y)}{9(2x + 1)},$$

$$0 < y < 3, \quad 0 < x < 1$$

(f)

$$\begin{aligned} P\left(1 < Y < 2 \mid X = \frac{1}{2}\right) &= \int_1^2 f_2\left(y \mid \frac{1}{2}\right) dy = \int_1^2 \frac{2 \left(\frac{3}{2} + y\right)}{9(1 + 1)} dy = \int_1^2 \left(\frac{y}{9} + \frac{1}{6}\right) dy \\ &= \left[\frac{y^2}{18} + \frac{y}{6} \right]_1^2 = \left(\frac{4}{18} + \frac{1}{3} \right) - \left(\frac{1}{18} + \frac{1}{6} \right) = \frac{1}{3} \end{aligned}$$

5.20

(a)

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^x \frac{1}{2} dy = \frac{y}{2} \Big|_0^x = \frac{x}{2}, \quad 0 < x < 2$$

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_y^2 \frac{1}{2} dx = \frac{x}{2} \Big|_y^2 = \frac{2-y}{2}, \quad 0 < y < 2$$

(b)

$$f_2(y|x) = \frac{f(x, y)}{f_1(x)} = \frac{1/2}{x/2} = \frac{1}{x}, \quad 0 < y < x < 2$$

(c)

$$f_1(x|y) = \frac{f(x, y)}{f_2(y)} = \frac{1/2}{(2-y)/2} = \frac{1}{2-y}, \quad 0 < y < x < 2$$

(d)

$$E(X) = \int_{-\infty}^{\infty} x f_1(x) dx = \int_0^2 \frac{x^2}{2} dx = \frac{x^3}{6} \Big|_0^2 = \frac{4}{3}$$

$$E(Y) = \int_{-\infty}^{\infty} y f_2(y) dy = \int_0^2 \left(y - \frac{y^2}{2} \right) dy = \left[\frac{y^2}{2} - \frac{y^3}{6} \right]_0^2 = 2 - \frac{4}{3} = \frac{2}{3}$$

(e)

$$E(Y|X = x) = \int_{-\infty}^{\infty} y f_2(y|x) dy = \int_0^x \frac{y}{x} dy = \frac{y^2}{2x} \Big|_0^x = \frac{x}{2}, \quad 0 < x < 2$$

$$E(X|Y = y) = \int_{-\infty}^{\infty} x f_1(x|y) dx = \int_y^2 \frac{x}{2-y} dx = \frac{x^2}{2(2-y)} \Big|_y^2 = \frac{4-y^2}{2(2-y)}$$

$$= \frac{(2-y)(2+y)}{2(2-y)} = \frac{y+2}{2}, \quad 0 < y < 2$$

(f)

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_1(x) dx = \int_0^2 \frac{x^3}{2} dx = \frac{x^4}{8} \Big|_0^2 = 2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 2 - \left(\frac{4}{3} \right)^2 = \frac{2}{9}$$

$$E(Y^2) = \int_{-\infty}^{\infty} y^2 f_2(y) dy = \int_0^2 \left(y^2 - \frac{y^3}{2} \right) dy = \left[\frac{y^3}{3} - \frac{y^4}{8} \right]_0^2 = \frac{8}{3} - 2 = \frac{2}{3}$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{2}{3} - \left(\frac{2}{3} \right)^2 = \frac{2}{9}$$

(g)

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x,y)dydx = \int_0^2 \int_0^x \frac{xy}{2} dydx = \int_0^2 \left(\frac{xy^2}{4} \Big|_{y=0}^x \right) dx = \int_0^2 \frac{x^3}{4} dx \\ &= \frac{x^4}{16} \Big|_0^2 = 1 \end{aligned}$$

$$Cov(X,Y) = E(XY) - E(X)E(Y) = 1 - \frac{4}{3} \cdot \frac{2}{3} = \frac{1}{9}$$

(h)

$$\rho = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{1/9}{2/9} = \frac{1}{2}$$

5.22

(a)

$$E(X) = \sum_{x=0}^3 xf(x) = 0(0.1) + 0.2 + 2(0.4) + 3(0.3) = 1.9$$

(b)

$$E(X^2) = \sum_{x=0}^3 x^2 f(x) = 0(0.1) + 0.2 + 4(0.4) + 9(0.3) = 4.5$$

$$Var(X) = E(X^2) - [E(X)]^2 = 4.5 - 1.9^2 = 0.89$$

$$\sigma = \sqrt{0.89} = 0.94$$

(c) $E(X^2 - 2) = E(X^2) - 2 = 4.5 - 2 = 2.5$

5.24

(a)

$$f_1(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad f_2(y) = \begin{cases} \frac{1}{3}e^{-y/3}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

By independence,

$$f(x,y) = f_1(x)f_2(y) = \begin{cases} \frac{1}{6}e^{-\frac{x}{2}-\frac{y}{3}}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(b) By independence,

$$f_2(y|x) = f_2(y) = \frac{1}{3}e^{-\frac{y}{3}}, y \geq 0$$

(c) By independence,

$$\begin{aligned} P(X > 3, Y > 4) &= P(X > 3)P(Y > 4) = [1 - F_1(3)][1 - F_2(4)] = e^{-3/2}e^{-4/3} \\ &= e^{-\frac{3}{2} - \frac{4}{3}} = e^{-\frac{17}{6}} = 0.0588 \end{aligned}$$

(d) By independence,

$$P(X > 3|Y = 3) = P(X > 3) = 1 - F_1(3) = e^{-3/2} = 0.2231$$

(e)

$$E(2X + Y) = 2E(X) + E(Y) = 2 \cdot 2 + 3 = 7$$

700 hours

(f)

$$\text{Var}(2X + Y) = 4\text{Var}(X) + \text{Var}(Y) = 4 \cdot 2^2 + 3^2 = 25$$

2500 hours

5.29

(a) $X_1 + \dots + X_4 \sim N(\mu_1 + \dots + \mu_4, \sigma_1^2 + \dots + \sigma_4^2) = N(40, 8)$

$$\begin{aligned} P(35 < X_1 + \dots + X_4 < 45) &= P\left(\frac{35 - 40}{\sqrt{8}} < Z < \frac{45 - 40}{\sqrt{8}}\right) \\ &= P(-1.768 < Z < 1.768) = \Phi(1.768) - \Phi(-1.768) \\ &= 0.9615 - 0.0385 = 0.923 \end{aligned}$$

(b)

$$\begin{aligned} \bar{X} &\sim N(10, 2/4) = N(10, 0.5) \\ P(\bar{X} < 11) &= P\left(Z < \frac{11 - 10}{\sqrt{0.5}}\right) = \Phi(1.414) = 0.9213 \end{aligned}$$

(c) $4X_1 + X_2 - X_3 - 4X_4 \sim N(4\mu_1 + \mu_2 - \mu_3 - 4\mu_4, 16\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 16\sigma_4^2) = N(0, 68)$

$$P(4X_1 + X_2 - X_3 - 4X_4 < 3) = P\left(Z < \frac{3}{\sqrt{68}}\right) = \Phi(0.364) = 0.6421$$

(d)

$$\begin{aligned} X_1 + \dots + X_4 &\sim N(\mu_1 + \dots + \mu_4, \sigma_1^2 + \dots + \sigma_4^2) = N(2 \cdot 8 + 2 \cdot 12, 2 \cdot 1 + 2 \cdot 2) \\ &= N(40, 6) \end{aligned}$$

$$\begin{aligned} P(35 < X_1 + \dots + X_4 < 45) &= P\left(\frac{35 - 40}{\sqrt{6}} < Z < \frac{45 - 40}{\sqrt{6}}\right) \\ &= P(-2.04 < Z < 2.04) = \Phi(2.04) - \Phi(-2.04) \\ &= 0.9793 - 0.0207 = 0.9586 \end{aligned}$$